Table 2. Total Global Bond Count

Molecule	Total Global Bond Count
Hydrogen Gas (H_2)	1
Sodium Chloride (NaCl)	1
Carbon dioxide (CO_2)	4
Water (H_2O)	2
Ammonia (NH_3)	3
Sulfur dioxide (SO_2)	3
Hydrogen sulfide (H_2S)	2
Phosphine (PH_3)	3
Uranium Dioxide (UO_2)	4
Lactic acid ($C_3H_6O_3$)	12
Dopamine ($C_8H_{11}NO_2$)	25
Insulin ($C_{257}H_{383}N_{65}O_{77}S_6$)	886
Cold Shock Protein (CspA) ($C_{335}H_{545}N_{91}O_{95}S_3$)	1177
Beta-lactoglobulin ($C_{817}H_{1285}N_{221}O_{251}S_8$)	2867
Hemoglobin ($C_{2952}H_{4664}N_{812}O_{832}S_8Fe_4$)	10300
DNA Polymerase I ($C_{4350}H_{6940}N_{1180}O_{1320}S_{40}$)	15300
Titin Fragment ($C_{5850}H_{9320}N_{1600}O_{1740}S_{54}$)	20554

Table 1 summarizes the total number of chemical bonds present in each molecular system examined in this study. The bond counts provide a simple structural measure of molecular complexity and serve as contextual reference information for the computational performance analysis discussed later in the manuscript. In the APEPM framework, the number of bonds within a molecule determines the total number of pairwise atomic overlap interactions that must be evaluated during a full molecular calculation. Consequently, molecules with larger bond networks correspond to a greater number of electron cloud overlap evaluations and increased computational workload. While the bond counts reported here are not directly used as parameters in the overlap calculations themselves, they provide a useful metric for interpreting differences in computational scaling and runtime behavior across the molecular systems considered in this work.

Table 8. Electron Cloud Radii Values, Ångströms (Å)

Elements	Electron Cloud Radii
H (Hydrogen)	1.30
He (Helium)	1.60
Li (Lithium)	2.05
Be (Beryllium)	1.75
B (Boron)	2.15
C (Carbon)	1.90
N (Nitrogen)	1.75
O (Oxygen)	1.72
F (Fluorine)	1.20
Ne (Neon)	1.75
Na (Sodium)	1.76
Mg (Magnesium)	1.85
Al (Aluminium)	2.10
Si (Silicon)	2.35
P (Phosphorus)	2.05
S (Sulfur)	2.05
Cl (Chlorine)	1.95
Ar (Argon)	2.10
K (Potassium)	2.55
Ca (Calcium)	2.38
Sc (Scandium)	2.25
Ti (Titanium)	2.35
V (Vanadium)	2.35
Cr (Chromium)	2.35
Mn (Manganese)	2.40
Fe (Iron)	2.40
Co (Cobalt)	2.82

Ni (Nickel)	1.85
Cu (Copper)	4.23
Zn (Zinc)	2.38
Ga (Gallium)	2.10
Ge (Germanium)	2.35
As (Arsenic)	2.10
Se (Selenium)	2.20
Br (Bromine)	2.05
Kr (Krypton)	2.25
Rb (Rubidium)	2.82
Sr (Strontium)	2.82
Y (Yttrium)	2.82
Zr (Zirconium)	1.60
Nb (Niobium)	2.55
Mo (Molybdenum)	2.55
Tc (Technetium)	2.60
Ru (Ruthenium)	2.60
Rh (Rhodium)	2.55
Pd (Palladium)	1.95
Ag (Silver)	5.32
Cd (Cadmium)	5.32
In (Indium)	2.20
Sn (Tin)	2.35
Sb (Antimony)	2.20
Te (Tellurium)	2.25
I (Iodine)	2.20
Xe (Xenon)	2.35
Cs (Cesium)	5.32

Ba (Barium)	5.32
La (Lanthanum)	2.40
Ce (Cerium)	2.55
Pr (Praseodymium)	2.55
Nd (Neodymium)	2.55
Pm (Promethium)	2.60
Sm (Samarium)	2.70
Eu (Europium)	2.75
Gd (Gadolinium)	2.80
Tb (Terbium)	2.85
Dy (Dysprosium)	2.90
Ho (Holmium)	2.90
Er (Erbium)	2.95
Tm (Thulium)	3.00
Yb (Ytterbium)	3.05
Lu (Lutetium)	3.10
Hf (Hafnium)	2.60
Ta (Tantalum)	2.60
W (Tungsten)	2.60
Re (Rhenium)	2.70
Os (Osmium)	2.70
Ir (Iridium)	2.70
Pt (Platinum)	1.95
Au (Gold)	5.32
Hg (Mercury)	5.32
Tl (Thallium)	2.20
Pb (Lead)	2.20
Bi (Bismuth)	2.30

Po (Polonium)	2.30
At (Astatine)	2.30
Rn (Radon)	2.40
Fr (Francium)	5.32
Ra (Radium)	5.32
Ac (Actinium)	5.32
Th (Thorium)	2.85
Pa (Protactinium)	2.85
U (Uranium)	2.85
Np (Neptunium)	2.95
Pu (Plutonium)	2.95
Am (Americium)	3.05
Cm (Curium)	3.05
Bk (Berkelium)	3.15
Cf (Californium)	3.15
Es (Einsteinium)	3.25
Fm (Fermium)	3.25
Md (Mendelevium)	3.25
No (Nobelium)	3.35
Lr (Lawrencium)	3.35
Rf (Rutherfordium)	3.00
Db (Dubnium)	3.00
Sg (Seaborgium)	3.00
Bh (Bohrium)	3.00
Hs (Hassium)	3.00
Mt (Meitnerium)	3.00
Ds (Darmstadtium)	3.00
Rg (Roentgenium)	3.00

Cn (Copernicium)	3.00
Nh (Nihonium)	3.00
Fl (Flerovium)	3.00
Mc (Moscovium)	3.00
Lv (Livermorium)	3.00
Ts (Tennessine)	3.00
Og (Oganesson)	3.00

Table 7 reports the effective electron cloud radius (ER) for each element across the periodic table, expressed in Ångströms (Å). In APEPM, the ER defines the outer radial boundary of an atom's probabilistic electron cloud, corresponding to the radial distance that encloses 95% of the atom's integrated ground-state electron probability density. These values serve as the primary geometric input parameters for all APEPM overlap calculations, specifically, the overlap radius (R), overlap area (Roa), and overlap volume (Rov). The ER for each element was derived from atomic radial probability distributions using a cumulative probability containment threshold of p = 0.95, and the resulting values fall within the same radial range as the $\rho = 0.001$ e/bohr^3 electron density isosurfaces commonly used in quantum-chemical molecular visualization, confirming their physical representativeness.

Table 9. Orbital $R_{i, min}$ And $R_{i, max}$ Values, Ångströms (Å)

Orbitals	**Orbital $R_{i, min}$ And $R_{i, max}$ Values**
1s	$R_{i, min} = 0.0$, $R_{i, max} = 0.53$
2s	$R_{i, min} = 0.53$, $R_{i, max} = 1.06$
2p	$R_{i, min} = 0.53$, $R_{i, max} = 1.20$
3s	$R_{i, min} = 1.06$, $R_{i, max} = 1.76$
3p	$R_{i, min} = 1.20$, $R_{i, max} = 2.38$
3d	$R_{i, min} = 1.76$, $R_{i, max} = 2.82$
4s	$R_{i, min} = 2.38$, $R_{i, max} = 4.23$
4p	$R_{i, min} = 2.82$, $R_{i, max} = 4.76$
4d	$R_{i, min} = 3.16$, $R_{i, max} = 5.32$

4f	$R_{i,min} = 3.16$, $R_{i,max} = 4.23$
5s	$R_{i,min} = 4.23$, $R_{i,max} = 6.61$
5p	$R_{i,min} = 4.76$, $R_{i,max} = 8.46$
5d	$R_{i,min} = 5.32$, $R_{i,max} = 9.49$
5f	$R_{i,min} = 5.32$, $R_{i,max} = 6.61$
5g	$R_{i,min} = 5.32$, $R_{i,max} = 6.61$
6s	$R_{i,min} = 6.61$, $R_{i,max} = 9.49$
6p	$R_{i,min} = 8.46$, $R_{i,max} = 12.98$
6d	$R_{i,min} = 9.49$, $R_{i,max} = 13.23$
6f	$R_{i,min} = 9.49$, $R_{i,max} = 12.00$
6g	$R_{i,min} = 9.49$, $R_{i,max} = 11.00$
6h	$R_{i,min} = 9.49$, $R_{i,max} = 10.00$
7s	$R_{i,min} = 12.98$, $R_{i,max} = 18.97$
7p	$R_{i,min} = 15.00$, $R_{i,max} = 25.96$
7d	$R_{i,min} = 15.00$, $R_{i,max} = 20.00$
7f	$R_{i,min} = 15.00$, $R_{i,max} = 18.00$
7g	$R_{i,min} = 15.00$, $R_{i,max} = 17.00$
7h	$R_{i,min} = 15.00$, $R_{i,max} = 16.00$
7i	$R_{i,min} = 17.00$, $R_{i,max} = 16.00$

Table 8 reports the inner ($R_{i,min}$) and outer ($R_{i,max}$) radial boundary values for each atomic orbital shell considered in APEPM, expressed in Ångströms (Å). These boundaries define the radial orbital shells used to determine orbital-level geometric participation in electron cloud overlap. For each orbital, the inner boundary corresponds to the nearest radial node or the distance at which probability density first exceeds 1% of the orbital's peak value, while the outer boundary is the distance at which probability density falls below 1% of its peak. An orbital is classified as geometrically participating in a bond when the overlap depth D = ER − R along the bond axis reaches or exceeds that orbital's $R_{i,min}$ boundary. These shell boundaries are not intended to represent quantum-mechanical eigenfunctions or energetic occupations;

rather, they serve as coarse-grained geometric partitions of electron probability support that enable APEPM to identify which orbital regions intersect the mandatory shared electron phase-space for any given bonded geometry.